2 Gaussian-chaos decomposition

We use the probabilists' Hermite polynomials

$$H_\alpha(G) = \prod_{I \in \mathcal{I}_N} \mathrm{He}_{\alpha_I}(g_I), \qquad \mathbb{E}[H_\alpha H_\eta] = \alpha!\, \mathbf{1}_{\{\alpha=\eta\}},$$

where $\alpha! = \prod_I \alpha_I!$. Since the H_α form a complete orthogonal basis of $L^2(\gamma_{\mathcal{I}_N})$, where $\gamma_{\mathcal{I}_N}$ denotes standard Gaussian measure on $\mathbb{R}^{\mathcal{I}_N}$, we have

$$Z_N(\beta : G) = \sum_{k=0}^{\infty} \sum_{\alpha:|\alpha|=k} \frac{\mathbb{E}[Z_N(\beta : G')H_\alpha(G')]}{\alpha!} H_\alpha(G), \tag{1}$$

where the second sum runs over all multi-index α such that $\sum_I \alpha_I = k$, and the expectation is taken over G' independent copy of G. Let P_r denote the orthogonal projection in $L^2(\gamma_{\mathcal{I}_N})$ onto the r-th homogeneous Gaussian chaos: Then, we can write

$$P_r Z_N(\beta; G) = \sum_{|\alpha|=r} \frac{\mathbb{E}[Z_N(\beta; G')H_\alpha(G')]}{\alpha!} H_\alpha(G).$$

Since $\mathbb{E}Z_N(\beta)$ is deterministic, we also have

$$P_r Y_N(\beta) = \frac{P_r Z_N(\beta)}{\mathbb{E}Z_N(\beta)}.$$

By the same orthogonality and Lemma 2.2, for every random variable $X = X(G)$,

$$\|P_r X\|_2^2 = \sum_{|\alpha|=r} \frac{|\mathbb{E}\,\partial_g^\alpha X|^2}{\alpha!},$$

whenever the expectations make senses.

Recall $A_N(\beta) := \frac{\beta}{2}\,\partial_\beta \log \mathbb{E}Z_N(\beta)$ from (**??**). The next lemma proves the exact radial decomposition

$$Y_N(\beta) - 1 = A_N(\beta)Q_N + \mathcal{R}_N(\beta), \qquad \mathcal{R}_N = \sum_{r\geq 3} P_r Y_N.$$

2.1 Exact finite-size radial projection

Call a multi-index α *even* if for every $x \in [N]$,

$$\sum_{I \ni x} \alpha_I \equiv 0 \pmod 2. \tag{2}$$

Otherwise, we call α odd.

Lemma 2.1. *For $s = (s_1, \dots, s_N) \in \{\pm 1\}^N$, define $\chi_s(I) = \prod_{x\in I} s_x$. We have*

$$Z_N(\beta; G) = Z_N\big(\beta; (\chi_s(I) g_I)_I\big). \tag{3}$$

In particular, if α is odd, then

$$\mathbb{E}[Z_N(\beta; G)H_\alpha(G)] = 0.$$

Proof. We define the Clifford volume element:

$$\Gamma := \psi_1 \cdots \psi_N.$$

Note that $\Gamma^*\Gamma$ is identity map, so Γ is unitary. Since N is even, moving ψ_j through the other $N-1$ Majorana generators produces the sign $(-1)^{N-1} = -1$. Hence, for any $j \in [N]$,

$$\Gamma\psi_j\Gamma^* = -\psi_j\Gamma\Gamma^* = -\psi_j.$$

For each $x \in [N]$, consider a single-coordinate reflection

$$U_x := \Gamma\psi_x.$$

This is unitary because $U_x^*U_x = \Gamma\psi_x\psi_x^*\Gamma^* = \mathbb{I}$. The Clifford relations (**??**) give

$$\psi_x\psi_j\psi_x = \begin{cases} \psi_x, & j = x, \\ -\psi_j, & j \neq x. \end{cases}$$

Hence, we obtain

$$U_x\psi_jU_x^* = \begin{cases} -\psi_x, & j = x, \\ \psi_j, & j \neq x. \end{cases} \tag{4}$$

Let $S_s := \{x \in [N] : s_x = -1\}$ and let U_s be the product of the unitaries U_x over $x \in S_s$:

$$U_s := \prod_{x \in S_s} U_x,$$

where the product is in any fixed order. Applying (4) successively yields that for any $j \in [N]$,

$$U_s\psi_jU_s^* = s_j\psi_j.$$

Recall that $\psi_I = \mathrm{i}^{q/2}\psi_{i_1}\cdots\psi_{i_q}$ for $I = \{i_1 < \cdots < i_q\}$. Hence, the above identity yields

$$\begin{aligned} U_s\psi_IU_s^* &= \mathrm{i}^{q/2}(U_s\psi_{i_1}U_s^*)\cdots(U_s\psi_{i_q}U_s^*) \\ &= \left(\prod_{x \in I} s_x\right)\psi_I = \chi_s(I)\psi_I. \end{aligned}$$

Writing $g^{(s)} = (\chi_s(I)g_I)_I$, we therefore have

$$\begin{aligned} H_N(g^{(s)}) &= \sigma_N \sum_{I \in \mathcal{I}_N} \chi_s(I)g_I\psi_I \\ &= \sigma_N \sum_{I \in \mathcal{I}_N} g_IU_s\psi_IU_s^* = U_sH_N(G)U_s^*. \end{aligned}$$

Using the unitary equivariance of the matrix exponential, we have

$$e^{-\beta H_N(g^{(s)})} = U_se^{-\beta H_N(G)}U_s^*.$$

Finally, since the trace is invariant under unitary conjugation, we conclude

$$Z_N(\beta; g^{(s)}) = \mathrm{tr}\Big(U_se^{-\beta H_N(G)}U_s^*\Big) = \mathrm{tr}\, e^{-\beta H_N(G)} = Z_N(\beta; G),$$

which proves (3).

Assume that α is odd so that we can take x such that $\sum_{I:x\in I}\alpha_I$ is odd. We take s such that $s_y = 1$ for $y \neq x$ and $s_x = -1$. Then,

$$H_\alpha((\chi_s(I)g_I)_I) = \left(\prod_{x=1}^N s_x^{\sum_{I\ni x}\alpha_I}\right)H_\alpha(G) = -H_\alpha(G).$$

Hence, we have

$$\mathbb{E}[Z_N(G')H_\alpha(G')] = \mathbb{E}[Z_N(\chi_sG')H_\alpha(\chi_sG')] = -\mathbb{E}[Z_N(G')H_\alpha(G')],$$

which gives the second claim. □

2.2 Gaussian integration by parts

Lemma 2.2. *Let $f : \mathbb{R}^{\mathcal{I}_N} \to \mathbb{R}$ be of class $C^{|\alpha|}$, and suppose that f and its mixed partial derivatives up to order α have at most exponential growth in $\|G\|$. Then*

$$\mathbb{E}[H_\alpha(G) f(G)] = \mathbb{E}\big[\partial_g^\alpha f(G)\big]\,. \tag{5}$$

In particular, (5) holds for $f(G) = Z_N(\beta; G)$.

Proof. Recall that the one-dimensional integration-by-parts identity yields

$$\mathbb{E}[\mathrm{He}_k(G) f(G)] = \mathbb{E}\Big[f^{(k)}(G)\Big]\,,$$

for every function $f \in C^k(\mathbb{R})$ satisfying the stated growth condition. Applying this identity successively to the independent Gaussian coordinates $(g_I)_{I\in\mathcal{I}_N}$ and using

$$H_\alpha(G) = \prod_{I\in\mathcal{I}_N} \mathrm{He}_{\alpha_I}(g_I),$$

gives

$$\begin{aligned}
\mathbb{E}[H_\alpha(G) f(G)] &= \mathbb{E}\left[\left(\prod_{I\in\mathcal{I}_N} \mathrm{He}_{\alpha_I}(g_I)\right) f(G)\right] \\
&= \mathbb{E}\left[\left(\prod_{I\in\mathcal{I}_N} \partial_{g_I}^{\alpha_I}\right) f(G)\right] \\
&= \mathbb{E}\big[\partial_g^\alpha f(G)\big]\,.
\end{aligned}$$

The growth assumption justifies all integrations by parts. Finally, for fixed N and β, the partition function $Z_N(\beta; G)$ is a smooth function of finitely many Gaussian coordinates, and all of its derivatives have at most exponential growth in $\|G\|$. Thus it satisfies the assumption. □

2.3 First and second moment

Lemma 2.3. *For every $\beta \in \mathbb{R}$,*

$$P_1 Y_N(\beta) = 0$$

and

$$P_2 Y_N(\beta) = \frac{\beta\,\partial_\beta \log \mathbb{E} Z_N(\beta)}{2M_N} \sum_{I\in\mathcal{I}_N} (g_I^2 - 1). \tag{6}$$

Proof. For multi-index α with $|\alpha| = 1$, since α is always odd, the first claim follows from Lemma 2.1.

In the second chaos it also kills every monomial $g_I g_J$ with $I \neq J$, since the corresponding multi-index is also odd. Consequently, since $\mathrm{He}_2(x) = x^2 - 1$,

$$P_2 Z_N(\beta) = \frac{1}{2} \sum_I \mathbb{E}[Z_N(\beta; G')((g_I')^2 - 1)](g_I^2 - 1).$$

It remains to compute the coefficient

$$b_N(\beta) := \frac{1}{2}\mathbb{E}[Z_N(\beta; G)(g_I^2 - 1)].$$

Applying Gaussian integration by parts yields

$$\mathbb{E}\big[Z_N(\beta)(g_I^2 - 1)\big] = \mathbb{E}\, g_I \partial_{g_I} Z_N(\beta).$$

On the other hand, since $Z_N(\beta; tG) = Z_N(t\beta; G)$, differentiating at $t = 1$ gives

$$\sum_I g_I \partial_{g_I} Z_N(\beta; G) = \beta\, \partial_\beta Z_N(\beta; G).$$

Hence, we have

$$2M_N b_N(\beta) = \sum_I \mathbb{E}\big[g_I \partial_{g_I} Z_N(\beta)\big] = \beta\, \partial_\beta \mathbb{E} Z_N(\beta).$$

This proves (6). □

2.4 Direct Rademacher control of all higher chaoses

Lemma 2.4. *For any $\beta \in \mathbb{R}$ and for any multi-index α with $|\alpha| = r$ satisfies*

$$\big|\partial_g^\alpha Z_N(\beta; G)\big| \le (|\beta|\sigma_N)^r Z_N(\beta; G). \tag{7}$$

Proof. The case $r = 0$ is immediate, so assume $r \ge 1$. Write $H_N = H_N(G)$ for brevity. We first derive the equality

$$\begin{aligned}
&\partial_{g_{I_r}} \cdots \partial_{g_{I_1}} Z_N(\beta; G) \\
&\quad = (-\beta\sigma_N)^r \sum_{\pi \in S_r} \int_{\Delta_r} \mathrm{tr}\Big(e^{-t_0\beta H_N} \psi_{I_{\pi(1)}} e^{-t_1\beta H_N} \cdots \psi_{I_{\pi(r)}} e^{-t_r \beta H_N}\Big)\, \mathrm{d}t_1 \cdots \mathrm{d}t_r
\end{aligned} \tag{8}$$

where

$$\Delta_r = \left\{ (t_j)_{j=1}^r : t_j \ge 0, \quad \sum_{j=1}^r t_j \le 1 \right\}, \qquad t_0 = 1 - \sum_{i=1}^r t_j.$$

For a differentiable matrix-valued function $A(\theta)$, the one-variable Duhamel formula is

$$\frac{\mathrm{d}}{\mathrm{d}\theta} e^{A(\theta)} = \int_0^1 e^{sA(\theta)} A'(\theta) e^{(1-s)A(\theta)}\, \mathrm{d}s. \tag{9}$$

By the definition of H_N in Subsection **??**,

$$\partial_{g_I} H_N = \sigma_N \psi_I.$$

Applying (9) to $A = -t\beta H_N$ gives, for $t \ge 0$,

$$\begin{aligned}
\partial_{g_I} e^{-t\beta H_N} &= -t\beta\sigma_N \int_0^1 e^{-st\beta H_N} \psi_I e^{-(1-s)t\beta H_N}\, \mathrm{d}s \\
&= -\beta\sigma_N \int_0^t e^{-u\beta H_N} \psi_I e^{-(t-u)\beta H_N}\, \mathrm{d}u.
\end{aligned} \tag{10}$$

Here, (10) with $t = 1$ yields (8) for $r = 1$. Suppose inductively that the formula holds at order $r - 1$. Then

$$\begin{aligned}
&\partial_{g_{I_r}} \cdots \partial_{g_{I_1}} Z_N(\beta; G) \\
&\quad = (-\beta\sigma_N)^{r-1} \sum_{\pi \in S_{r-1}} \int_{\Delta_{r-1}} \partial_{g_{I_r}} \mathrm{tr}\Big(e^{-t_0\beta H_N} \psi_{I_{\pi(1)}} e^{-t_1\beta H_N} \cdots \psi_{I_{\pi(r-1)}} e^{-t_{r-1}\beta H_N}\Big)\, \mathrm{d}t_1 \cdots \mathrm{d}t_{r-1}.
\end{aligned} \tag{11}$$

For fixed $\pi \in S_{r-1}$ and $j \in \{1, \dots, r\}$, differentiating the factor $e^{-t_{j-1}\beta H_N}$ replaces it, by (10), with

$$-\beta\sigma_N \int_0^{t_{j-1}} e^{-u\beta H_N} \psi_{I_r} e^{-(t_{j-1}-u)\beta H_N}\, \mathrm{d}u.$$

Splitting t_{j-1} into u and $t_{j-1}-u$ and relabeling the resulting simplex variables, the corresponding term in the Leibniz rule becomes

$$-\beta\sigma_N \int_{\Delta_r} \operatorname{tr}\Big(e^{-t_0\beta H_N}\psi_{I_{\sigma_{\pi,j}(1)}}e^{-t_1\beta H_N}\cdots\psi_{I_{\sigma_{\pi,j}(r)}}e^{-t_r\beta H_N}\Big)\,\mathrm{d}t_1\cdots\mathrm{d}t_r,$$

where

$$\sigma_{\pi,j}(k)=\begin{cases}\pi(k) & (k<j)\\ r & (k=j)\\ \pi(k-1) & (k>j)\end{cases}.$$

Since $(\pi,j)\mapsto\sigma_{\pi,j}$ is a bijection from $S_{r-1}\times\{1,\dots,r\}$ onto S_r, summing the Leibniz-rule terms in (11) proves (8).

It remains to estimate each integrand in (8). From now on the Schatten norms are defined using the normalized trace:

$$\|A\|_p=(\operatorname{tr}|A|^p)^{1/p},\qquad \|A\|_\infty=\|A\|\,.$$

For $t_j>0$,

$$\left\|e^{-t_j\beta H_N}\right\|_{1/t_j}=\left(\operatorname{tr}e^{-\beta H_N}\right)^{t_j},$$

with the usual $p=\infty$ interpretation when $t_j=0$. By Subsection **??**, $\|\psi_I\|_\infty=1$. Noncommutative Hölder, with Schatten exponents $1/t_j$, therefore gives, for every $\pi\in S_r$ and $t\in\Delta_r$,

$$\begin{aligned}&\left|\operatorname{tr}\Big(e^{-t_0\beta H_N}\psi_{I_{\pi(1)}}e^{-t_1\beta H_N}\cdots\psi_{I_{\pi(r)}}e^{-t_r\beta H_N}\Big)\right|\\ &\quad\le\prod_{j=0}^{r}\left\|e^{-t_j\beta H_N}\right\|_{1/t_j}\prod_{k=1}^{r}\left\|\psi_{I_{\pi(k)}}\right\|_\infty\\ &\quad\le\prod_{j=0}^{r}\left(\operatorname{tr}e^{-\beta H_N}\right)^{t_j}=\operatorname{tr}e^{-\beta H_N}=Z_N(\beta;G),\end{aligned}\tag{12}$$

where we used $\sum_{j=0}^{r}t_j=1$. Combining (8) and (12), we obtain

$$\begin{aligned}\left|\partial_{g_{I_r}}\cdots\partial_{g_{I_1}}Z_N(\beta;G)\right|&\le(|\beta|\sigma_N)^r\sum_{\pi\in S_r}\int_{\Delta_r}Z_N(\beta;G)\,\mathrm{d}t_1\cdots\mathrm{d}t_r\\ &=(|\beta|\sigma_N)^r Z_N(\beta;G)|S_r|\int_{\Delta_r}\mathrm{d}t_1\cdots\mathrm{d}t_r\\ &=(|\beta|\sigma_N)^r Z_N(\beta;G),\end{aligned}$$

which proves (7). □

Recall that Lemma 2.1 shows that only even multi-indices have nonzero Hermite coefficients.

Let $(\varepsilon_x)_{x=1}^N$ be independent Rademacher variables on an auxiliary probability space, and write $\mathbb{E}_\varepsilon$ for expectation on this space. Set

$$S_{N,q}=\sum_{I\in\mathcal{I}_N}\varepsilon_I,\qquad \varepsilon_I=\prod_{x\in I}\varepsilon_x.\tag{13}$$

Thus $S_{N,q}$ is the degree-q elementary symmetric polynomial in the Rademacher signs.

Lemma 2.5. *For every integer $r\ge0$,*

$$\sum_{\substack{|\alpha|=r\\ \alpha\text{ even}}}\frac{1}{\alpha!}=\frac{1}{r!}\mathbb{E}_\varepsilon\big[S_{N,q}^r\big]\,.\tag{14}$$

Moreover, there are constants $c_q > 0$ and $C_q < \infty$, depending only on q, such that

$$\mathbb{E}_\varepsilon\left[|S_{N,q}|^3 e^{t|S_{N,q}|}\right] \le C_q N^{3q/2} \tag{15}$$

for every N and every $0 \le t \le c_q N^{-(q-1)}$.

Proof. Starting from the definition (13) and applying the multinomial formula gives

$$\frac{1}{r!}\mathbb{E}_\varepsilon\left[S_{N,q}^r\right] = \frac{1}{r!}\mathbb{E}_\varepsilon\left[\sum_{|\alpha|=r} r! \prod_{I\in\mathcal{I}_N} \frac{\varepsilon_I^{\alpha_I}}{\alpha_I!}\right] = \sum_{|\alpha|=r} \frac{1}{\alpha!}\mathbb{E}_\varepsilon\left[\prod_{I\in\mathcal{I}_N} \varepsilon_I^{\alpha_I}\right]. \tag{16}$$

For a fixed α, the product inside the expectation is

$$\prod_{x=1}^{N} \varepsilon_x^{\sum_{I:x\in I}\alpha_I}.$$

By independence of the Rademacher variables and the definition of an even multi-index in (2), its expectation is one if α is even and zero otherwise. Substitution into (16) proves (14).

We next prove a deterministic bound for $S_{N,q}$. For every $N \ge 1$ and every integer $m \ge 0$, let $e_m^{(N)}$ denote the mth elementary symmetric polynomial in N variables, that is,

$$e_m^{(N)}(z_1,\dots,z_N) := \sum_{\substack{I\subset[N]\\|I|=m}} \prod_{x\in I} z_x, \qquad e_0^{(N)}(z_1,\dots,z_N) = 1.$$

For the rest of the proof, we use the abbreviation

$$e_m := e_m^{(N)}(\varepsilon_1,\dots,\varepsilon_N).$$

In particular, $e_1 = \sum_{x=1}^N \varepsilon_x$, and (13) gives $S_{N,q} = e_q$. Newton's identities state that

$$m e_m = \sum_{j=1}^{m} (-1)^{j-1} e_{m-j} p_j, \tag{17}$$

where $p_j := \sum_{x=1}^N \varepsilon_x^j$. Since $\varepsilon_x \in \{\pm 1\}$, we have

$$p_j = \begin{cases} e_1, & j \text{ odd},\\ N, & j \text{ even}. \end{cases}$$

An induction on m, using (17), shows that there is a constant $C_m < \infty$ such that

$$|e_m| \le C_m\left(N^{m/2} + |e_1|^m\right).$$

Indeed, suppose that the estimate is known below order m and set

$$X_N := \frac{e_1}{\sqrt{N}}.$$

For every $j \ge 1$,

$$|p_j| \le N^{j/2}(1 + |X_N|^j).$$

Hence, with $C_0 = 1$,

$$\begin{aligned} |e_{m-j}p_j| &\le C_{m-j} N^{m/2}(1+|X_N|^{m-j})(1+|X_N|^j)\\ &\le 2C_{m-j}N^{m/2}(1+|X_N|^m), \end{aligned}$$

where we used $1+|X|^m-|X|^{m-j}-|X|^j=(1-|X|^j)(1-|X|^{m-j})\geq 0$. Substitution into (17) proves the induction step. Taking $m=q$ and using (13), we obtain

$$|S_{N,q}|\leq C_q\left(N^{q/2}+|e_1|^q\right). \tag{18}$$

Since $q\geq 4$ by Subsection **??** and $|X_N|\leq\sqrt{N}$, (18) implies that, for $0\leq t\leq c_qN^{-(q-1)}$,

$$\begin{aligned} t|S_{N,q}| &\leq C_qc_qN^{-(q/2-1)}\left(1+|X_N|^q\right)\\ &\leq C_qc_q\left(1+|X_N|^2\right). \end{aligned} \tag{19}$$

Indeed,

$$N^{-(q/2-1)}|X_N|^q=|X_N|^2\left(\frac{|X_N|}{\sqrt{N}}\right)^{q-2}\leq|X_N|^2.$$

Choose c_q small enough that the last member of (19) is bounded by $(1+|X_N|^2)/4$.

Hoeffding's inequality gives

$$\mathbb{P}_\varepsilon(|X_N|\geq u)\leq 2e^{-u^2/2}\qquad(u\geq 0). \tag{20}$$

Consequently, integration of the tail in (20) yields

$$\sup_{N\geq 1}\mathbb{E}_\varepsilon\left[\left(1+|X_N|^{3q}\right)e^{|X_N|^2/4}\right]<\infty. \tag{21}$$

Finally, (18), (19), and (21) give

$$\begin{aligned} \mathbb{E}_\varepsilon\left[|S_{N,q}|^3e^{t|S_{N,q}|}\right] &\leq C_qN^{3q/2}\mathbb{E}_\varepsilon\left[\left(1+|X_N|^{3q}\right)e^{(1+|X_N|^2)/4}\right]\\ &\leq C_qN^{3q/2}, \end{aligned}$$

which proves (15). □

Proposition 2.6. *There exists $\beta_{\mathrm{Rad}}=\beta_{\mathrm{Rad}}(q)>0$ such that, for every fixed real β with $|\beta|<\beta_{\mathrm{Rad}}$,*

$$\sum_{r\geq 3}\|P_rY_N(\beta)\|_2^2=O\left(N^{-3(q/2-1)}\right)=o(N^{2-q}).$$

Proof. Put

$$t_N=\beta^2\sigma_N^2=\beta^2(q-1)!N^{-(q-1)}.$$

By the Hermite expansion (1) and its orthogonality relation, the squared norm of the rth projection is the sum of the squared Hermite coefficients. Lemma 2.1 eliminates every odd multi-index, while Lemma 2.2 replaces each remaining Hermite coefficient by the corresponding expected derivative. Consequently, Lemma 2.4 and $\mathbb{E}Z_N(\beta)>0$ give

$$\begin{aligned} \|P_rY_N(\beta)\|_2^2 &= \frac{1}{\mathbb{E}Z_N(\beta)^2}\sum_{\substack{|\alpha|=r\\ \alpha\text{ even}}}\frac{1}{\alpha!}\left|\mathbb{E}\partial_g^\alpha Z_N(\beta)\right|^2\\ &\leq t_N^r\sum_{\substack{|\alpha|=r\\ \alpha\text{ even}}}\frac{1}{\alpha!}. \end{aligned} \tag{22}$$

By the first assertion of Lemma 2.5, namely (14), summing (22) over all $r\geq 3$ yields

$$\begin{aligned} \sum_{r\geq 3}\|P_rY_N(\beta)\|_2^2 &\leq \sum_{r\geq 3}\frac{t_N^r}{r!}\mathbb{E}_\varepsilon\left[S_{N,q}^r\right]\\ &\leq \mathbb{E}_\varepsilon\left[\sum_{r\geq 3}\frac{(t_N|S_{N,q}|)^r}{r!}\right]\\ &\leq \frac{t_N^3}{6}\mathbb{E}_\varepsilon\left[|S_{N,q}|^3e^{t_N|S_{N,q}|}\right]. \end{aligned} \tag{23}$$

Here the second inequality uses $|\mathbb{E}_\varepsilon[S_{N,q}^r]| \le \mathbb{E}_\varepsilon[|S_{N,q}|^r]$, and the last one uses $\sum_{r\ge 3} u^r/r! \le u^3 e^u/6$ for $u \ge 0$. Choose

$$\beta_{\mathrm{Rad}} = \sqrt{\frac{c_q}{(q-1)!}},$$

where c_q is the constant in Lemma 2.5. If $|\beta| < \beta_{\mathrm{Rad}}$, then $t_N \le c_q N^{-(q-1)}$. Therefore the second assertion of Lemma 2.5, namely (15), together with (23) imply

$$\begin{aligned}\sum_{r\ge 3} \|P_r Y_N(\beta)\|_2^2 &\le C_{q,\beta} N^{-3(q-1)} N^{3q/2} \\ &= C_{q,\beta} N^{-3(q/2-1)}.\end{aligned}$$

Since $q \ge 4$, we have $3(q/2-1) > q-2$, which completes the proof. □